Create an AWS Application Load Balancer (ALB)

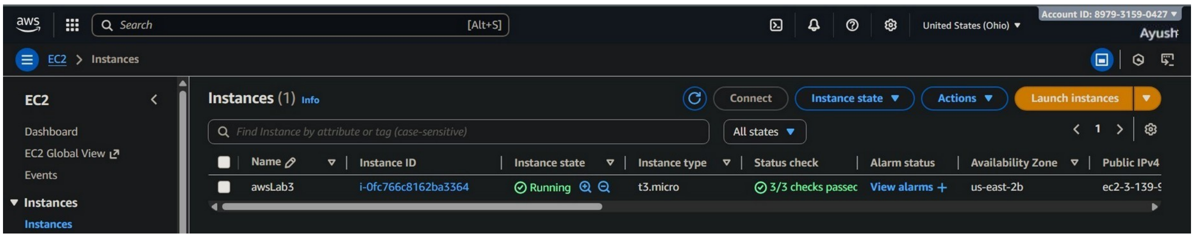
Create an Application Load Balancer that distributes incoming HTTP traffic across multiple EC2 instances.

Prerequisites

- AWS account with necessary permissions (EC2, ELB access)
- At least two running EC2 instances in the same VPC with a security group allowing HTTP (port 80)
- Basic understanding of AWS Console navigation

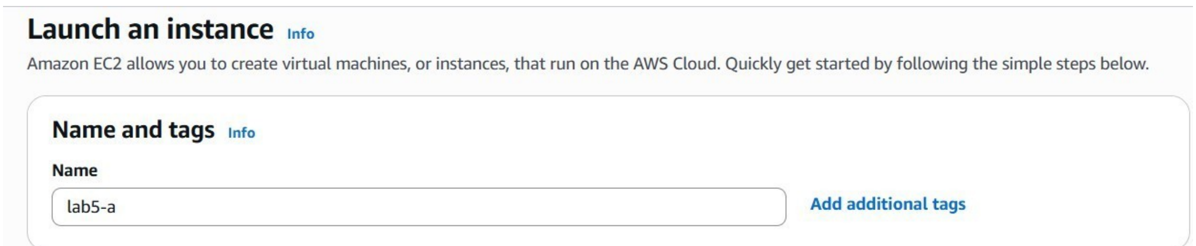
Step 1: Login and prepare environment

1. Login to your AWS Management Console at https://aws.amazon.com/console/
2. Navigate to **EC2 Dashboard** from the Services menu.

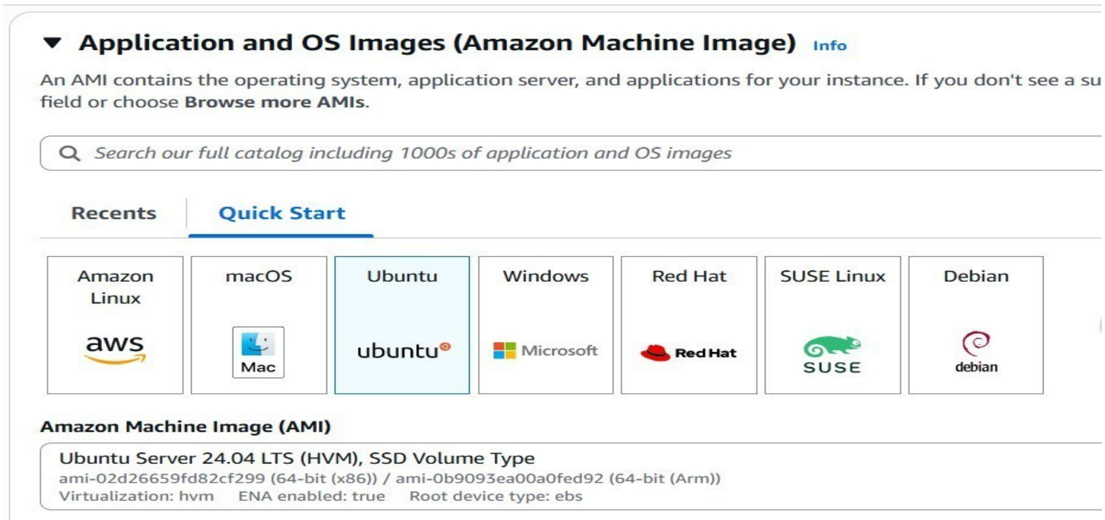


Step 2: Create Two new EC2 instances for ALB(Application Load Balancer) us

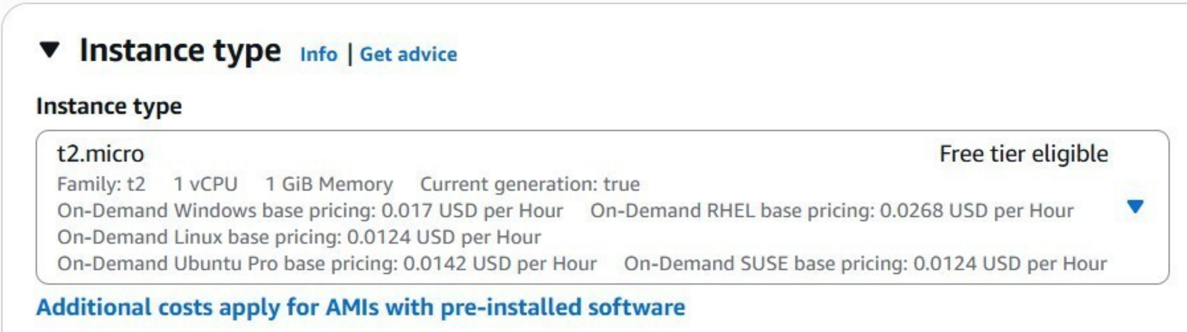
- Click on **LAUNCH INSTANCES.**
- **Fill the required data.**



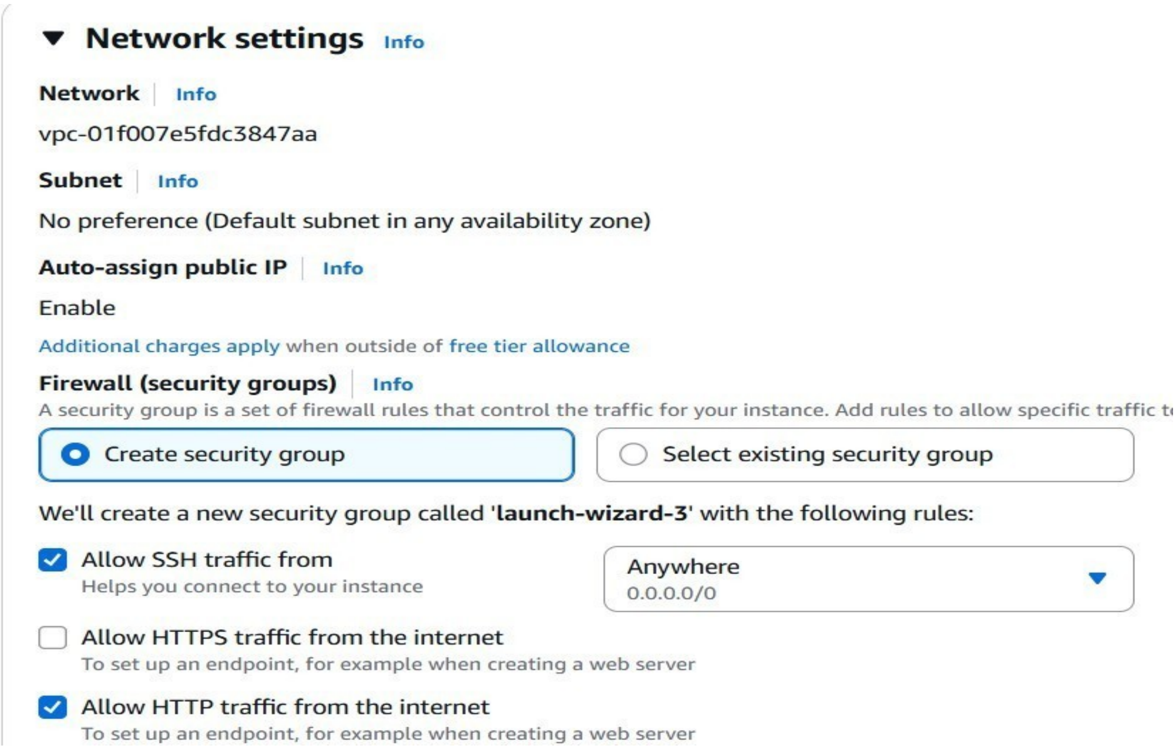
- **Choose UBUNTU as AMI**



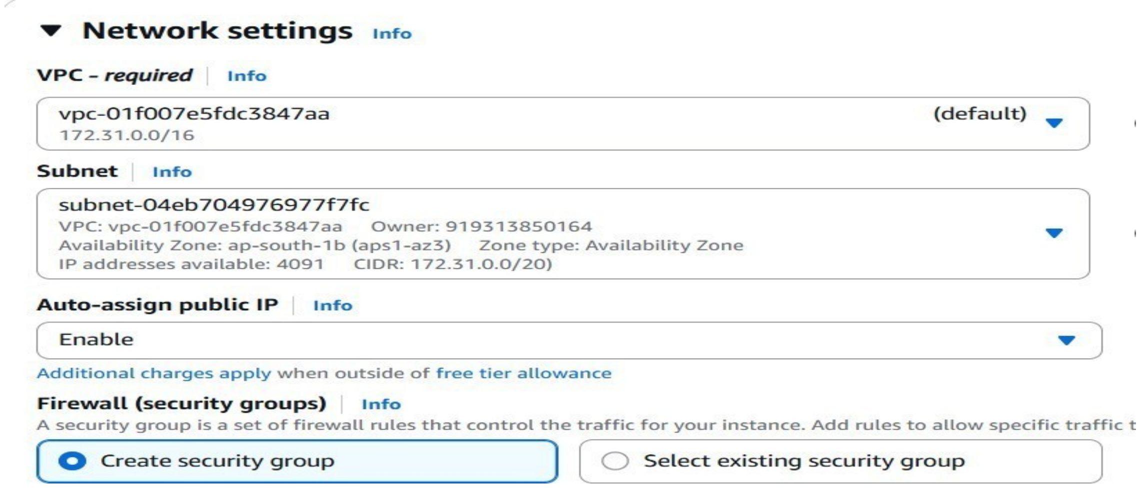
- Select the Instance Type (**free tier Eligible**).



- **In Network Settings** Allow **HTTP traffic** from the internet.



- Edit the Network Settings.



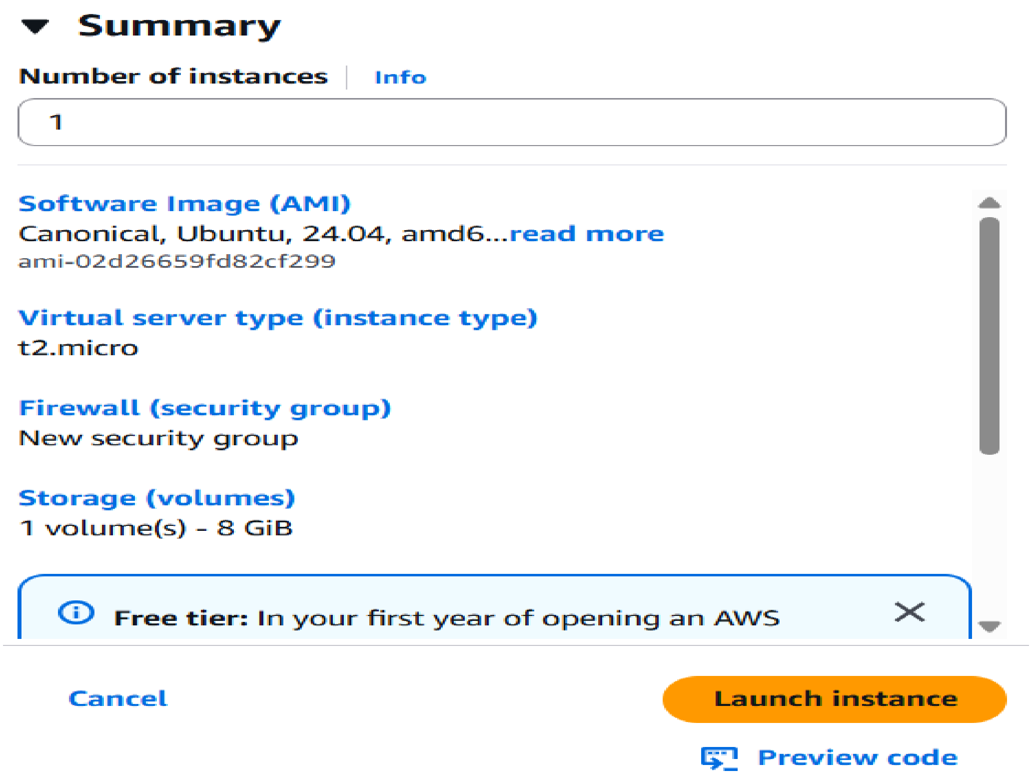
- **Visit Advanced Details.**
- Go to **User Data- optional** at the end of page.
- Paste the data in **WORKSPACE** or create a bin file with same data and Upload it.

    o #!/bin/bash

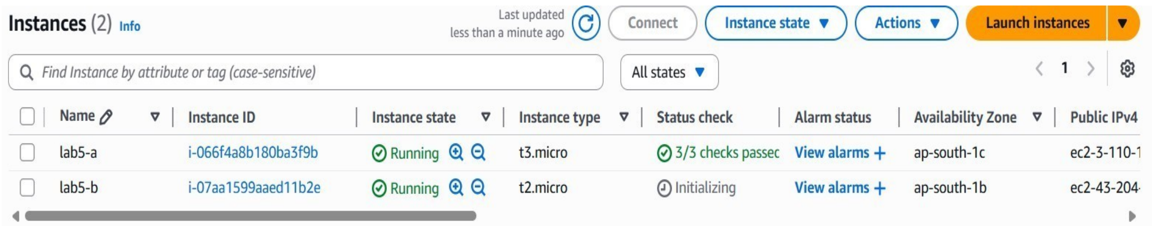
    # Update package lists
    o apt-get update

    # Install nginx without asking for confirmation
    o apt-get install nginx -y

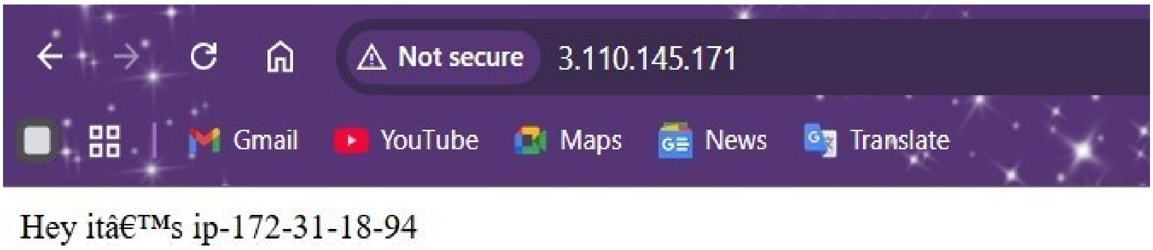
    # Set homepage to show "Hey it’s <hostname>"
    o echo "Hey it’s $(hostname)" > /var/www/html/index.html
- **Launch** the instance.

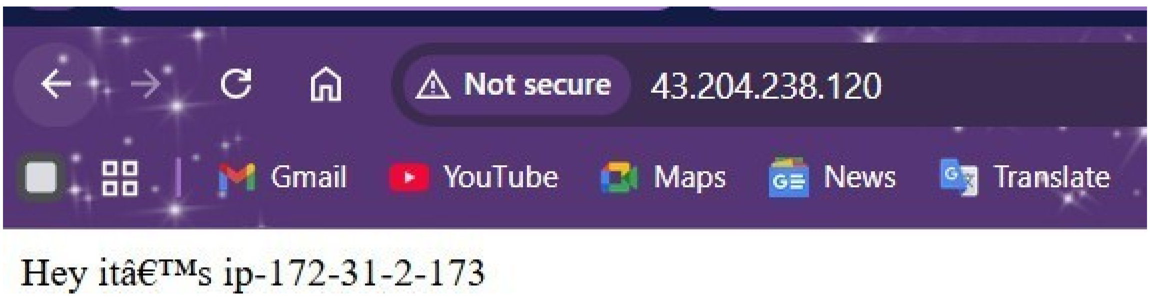


- **Create 2 similar instances.**



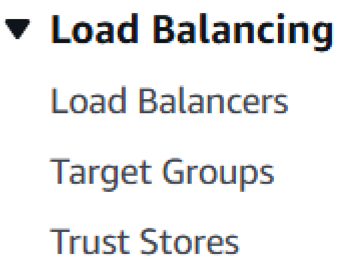
- **Run the public Ips of two instance in new tabs.**



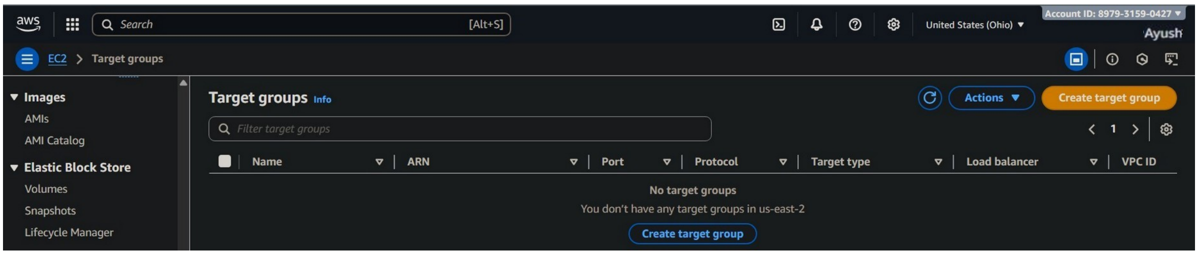


Step 3: Create a Target Group

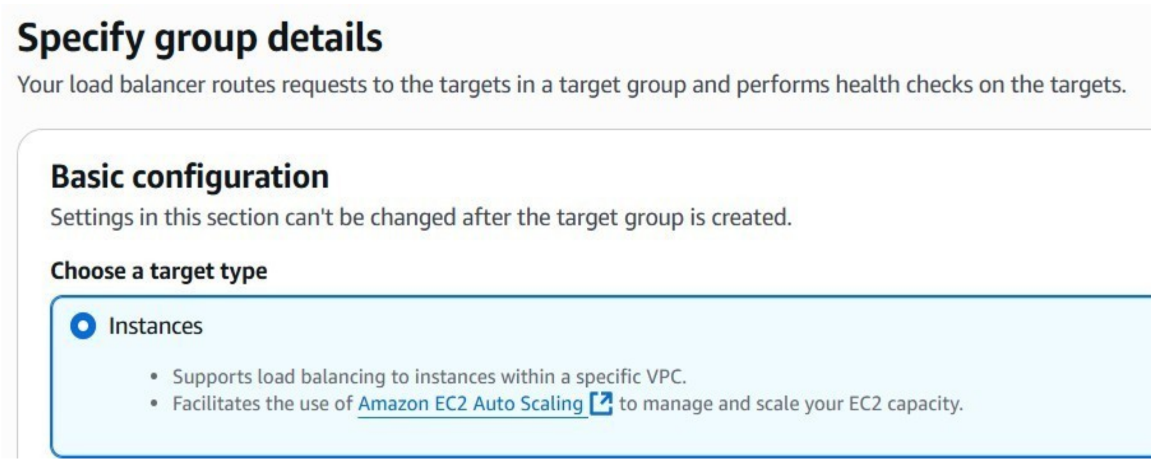
1. In the EC2 dashboard sidebar, locate **Load Balancing** and click **Target Groups**.



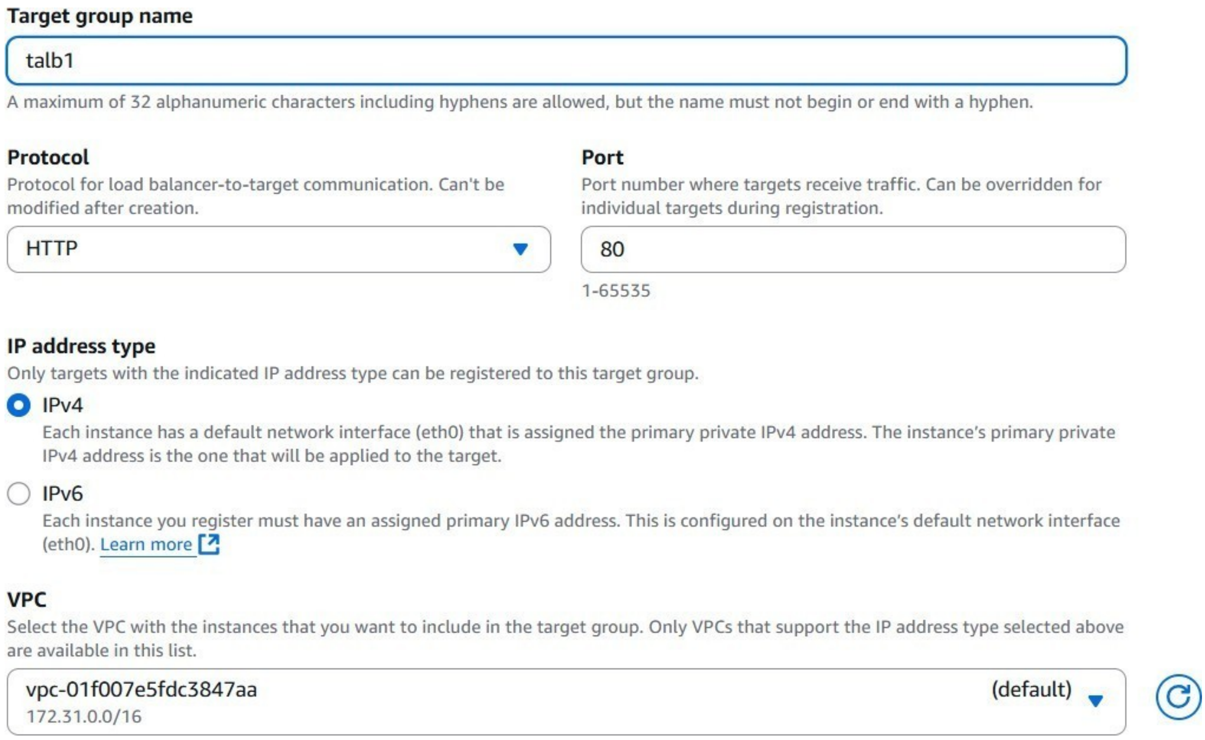
2. Click **Create target group**.



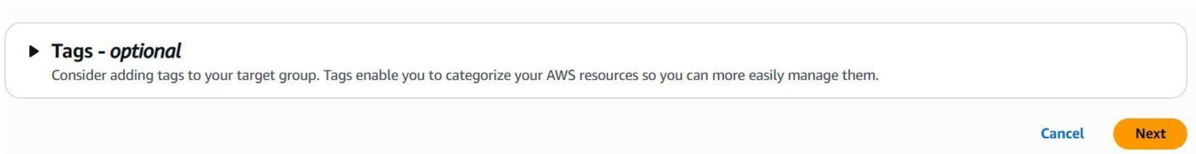
3. Select **Target type** as **Instance**.



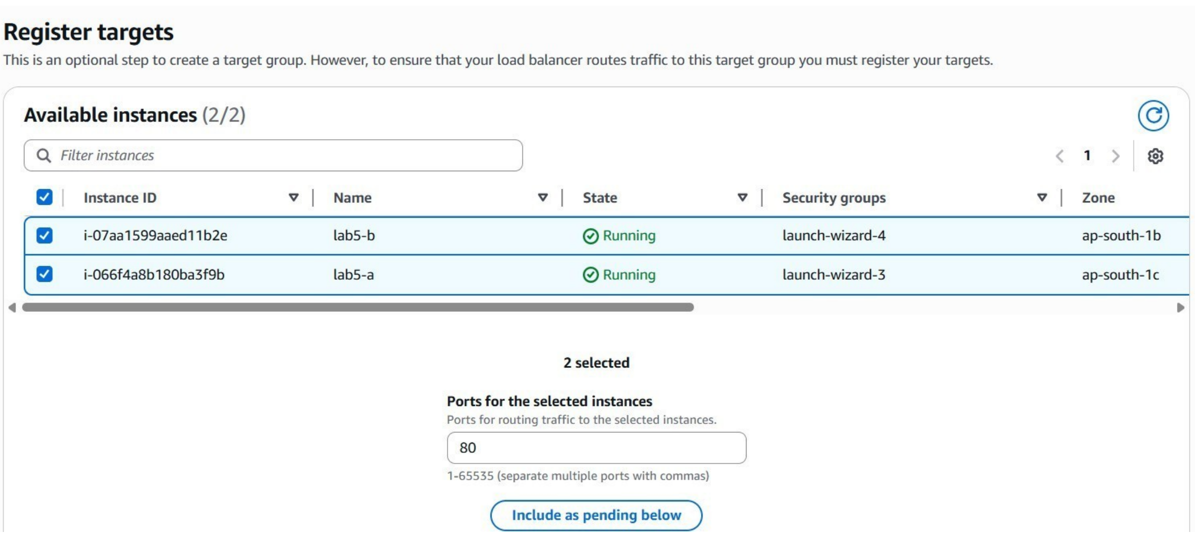
4. Enter a **Name** for your target group, e.g., MyTargetGroup.
5. Set **Protocol** to HTTP and **Port** to 80.
6. Select the correct **VPC** where your EC2 instances are located.



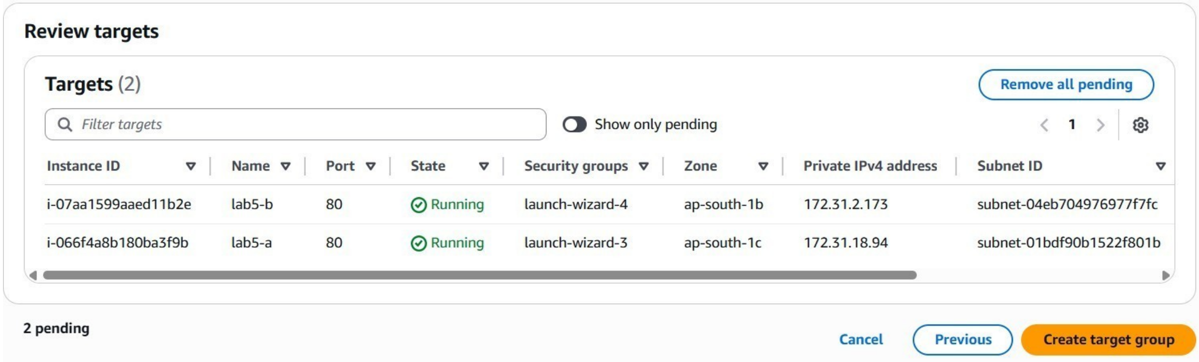
7. Click **Next**.



8. Under **Registered targets**, select your two EC2 instances.



9. Click **Add to registered** > then click **Create target group**.



Review targets

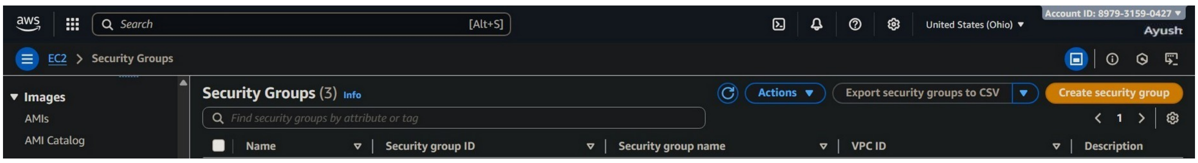
Targets (2)

Instance ID	Name	Port	State	Security groups	Zone	Private IPv4 address	Subnet ID
i-07aa1599aaed11b2e	lab5-b	80	Running	launch-wizard-4	ap-south-1b	172.31.2.173	subnet-04eb704976977f7fc
i-066f4a8b180ba3f9b	lab5-a	80	Running	launch-wizard-3	ap-south-1c	172.31.18.94	subnet-01bdf90b1522f801b

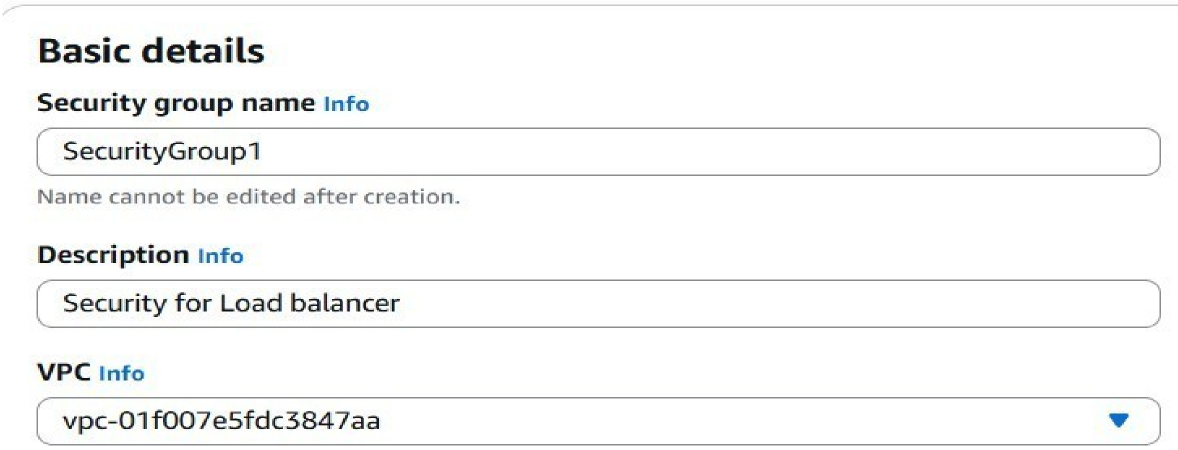
Step 4: Create a SECURITY GROUP

1. In the **EC2** Dashboard Sidebar, Visit **Network & Security**.

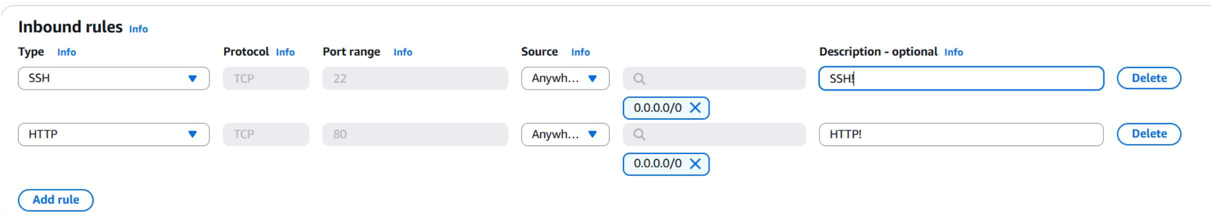
2. Click on **Create Security Group.**



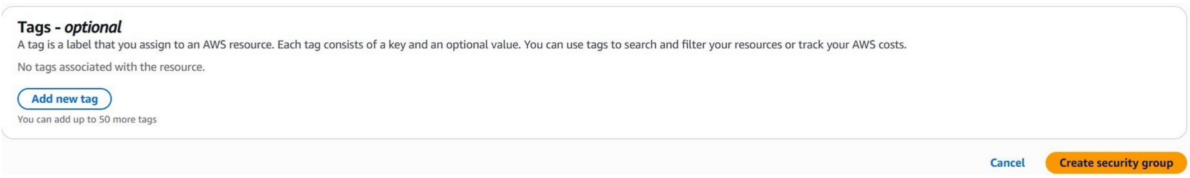
3. Add Security group name.
4. Fill the description-optional.
5. Select the correct **VPC.**



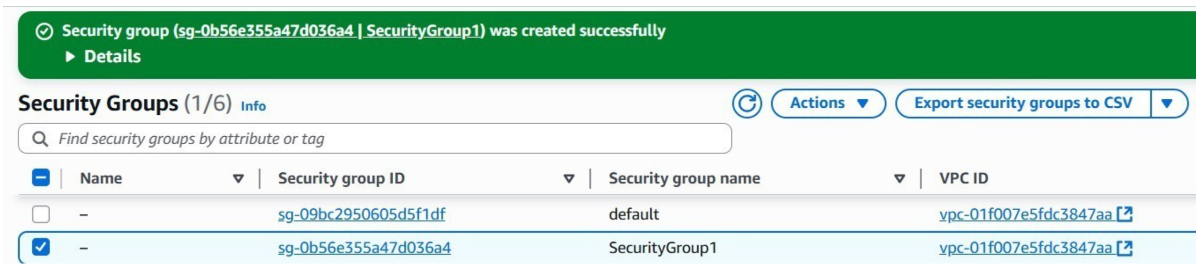
6. **ADD INBOUND RULES for SSH & HTTP** with source as **Anywhere-IPv4.**



7. Click on **Create security group.**



8. Verify the Security Group.



Step 5: Create the Application Load Balancer

1. In the EC2 dashboard sidebar, click **Load Balancers**.

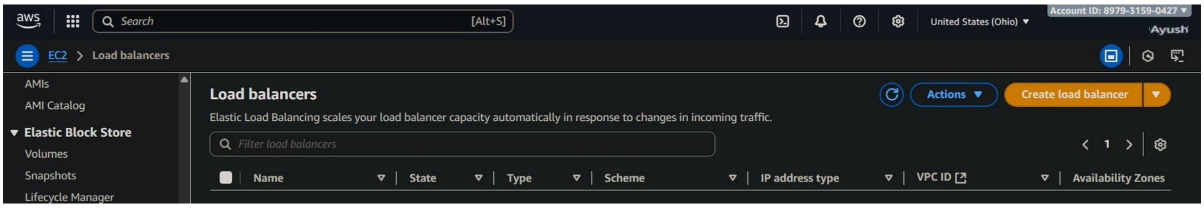
▼ **Load Balancing**

Load Balancers

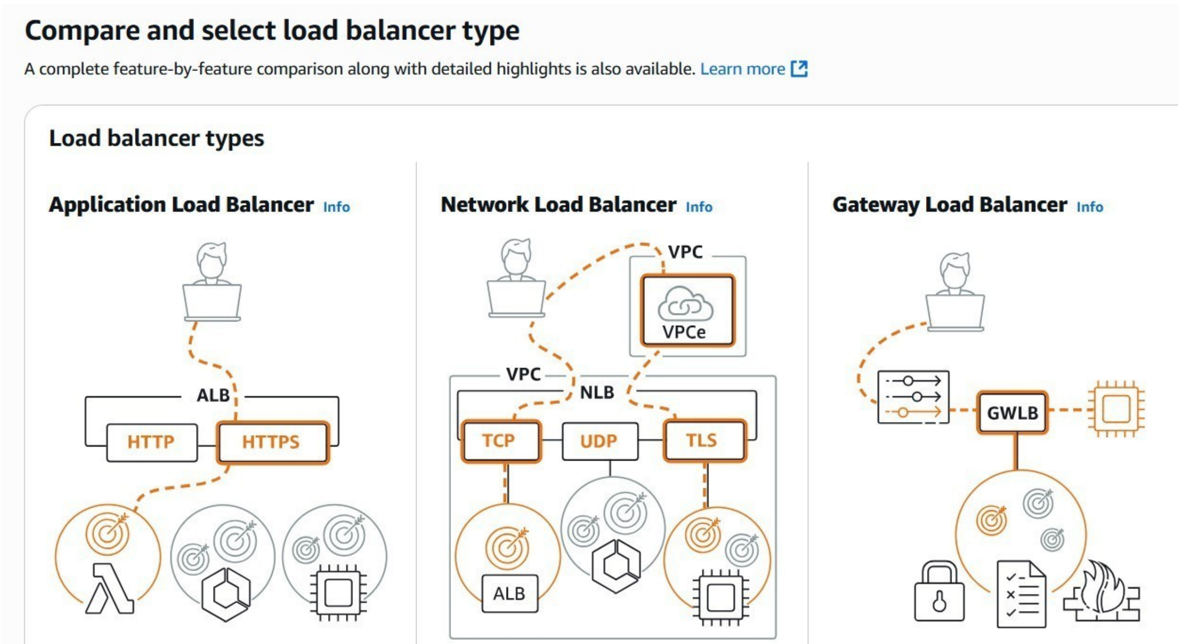
Target Groups

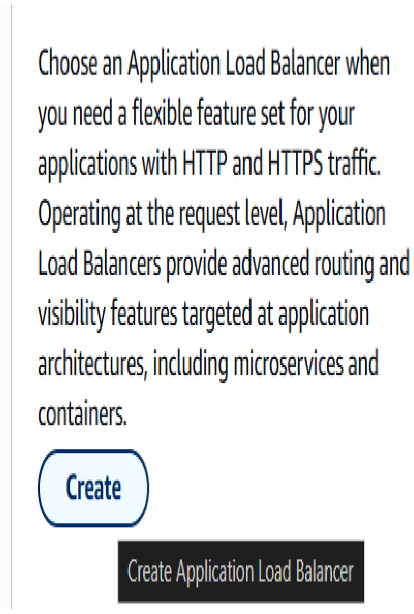
Trust Stores

2. Click **Create Load Balancer**.

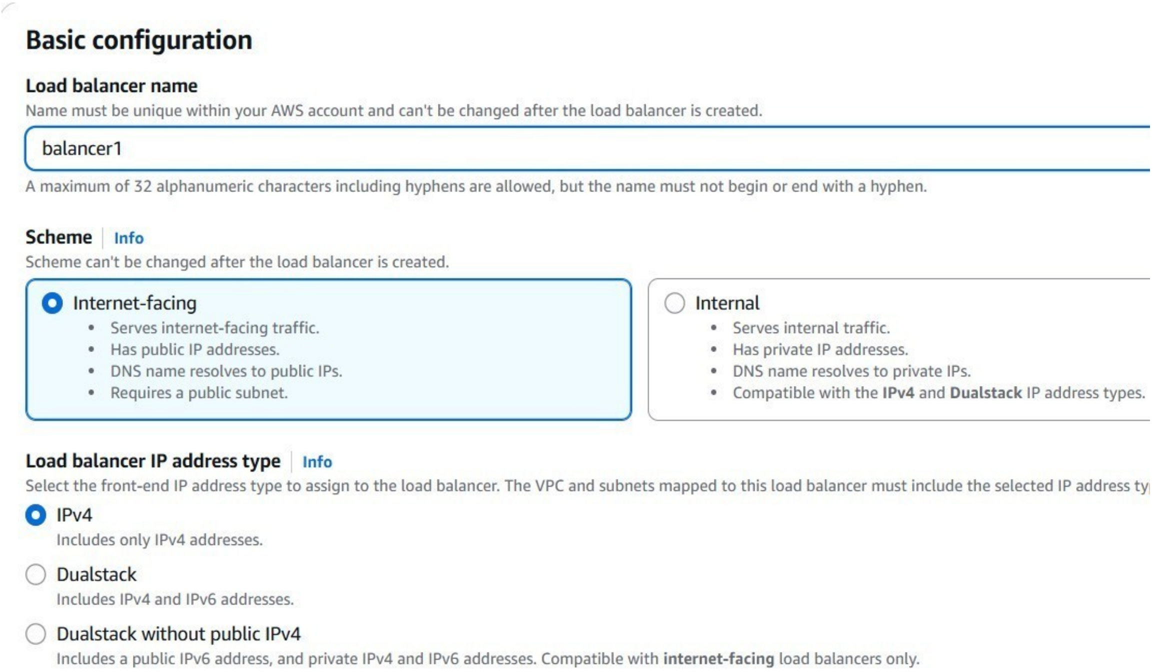


3. Select **Application Load Balancer** and click **Create**.

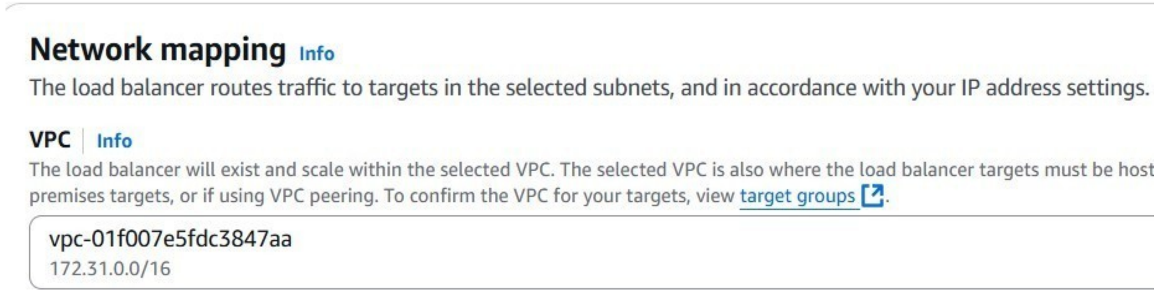


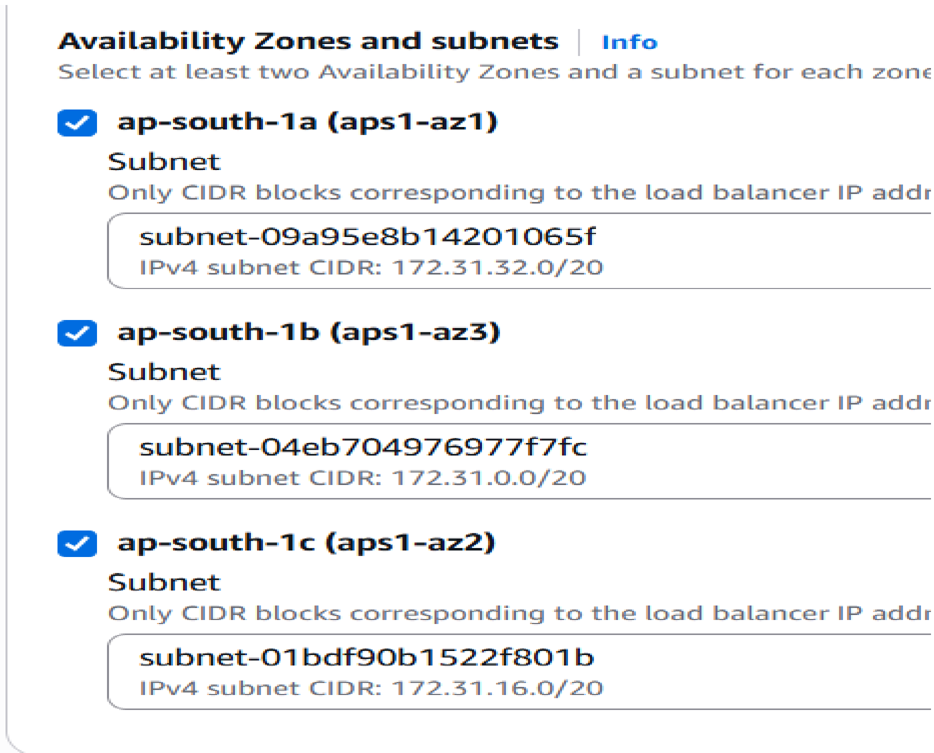


4. Enter a **Name** for the load balancer, e.g., MyAppLoadBalancer.
5. Set **Scheme** to **internet-facing**.
6. Choose **IPv4** as IP address type.

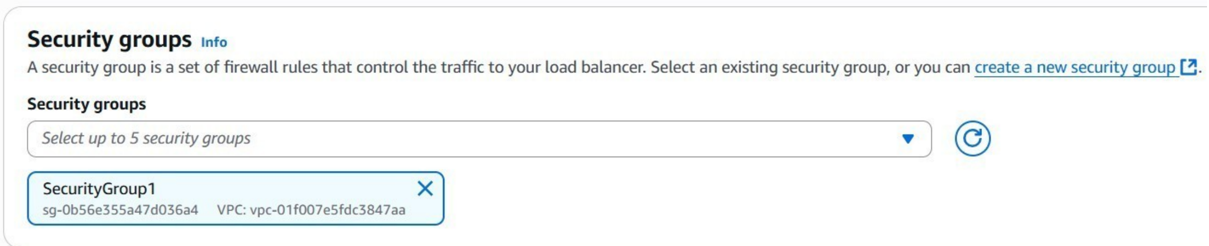


7. Choose the **VPC** and select at least two Availability Zones (AZ) subnets.

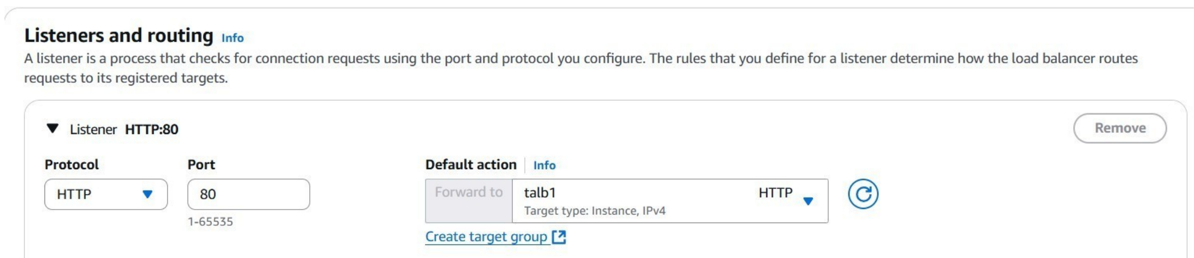




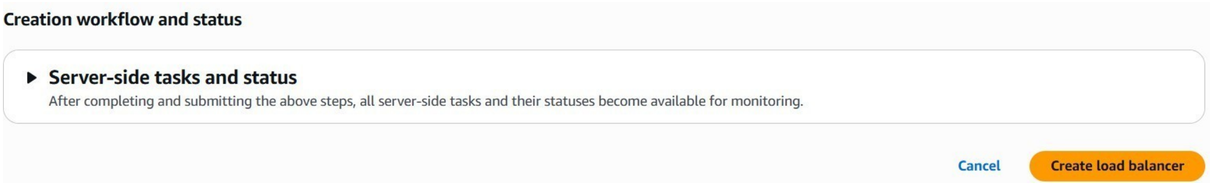
8. **Select the Security Group** (if created) **or create a new security group.**



9. Select the **Listeners and Routing**.

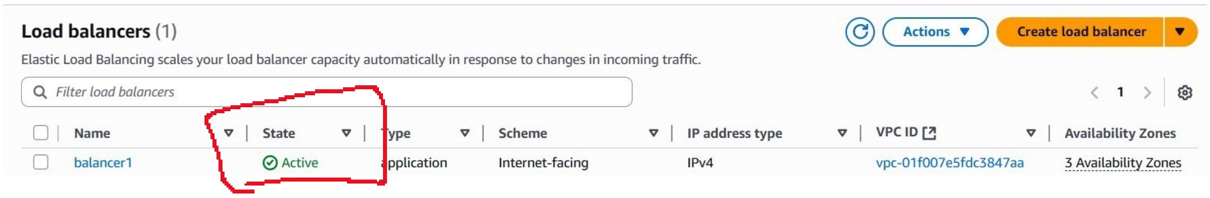


10.Have a quick **Review** and click on **Create load balancer**.

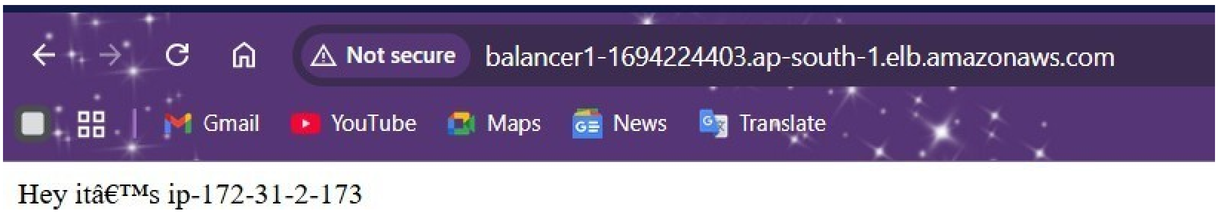


Step 6: Validate and Test

1. In the **Load Balancer List**, select the load balancer you just created.
2. Confirm the status changes from *provisioning* to *active*. When the Status is Active then only run the **DNS.**



3. Check the **Target Groups** tab and ensure the EC2 instances show as **healthy**.
4. Copy the **DNS name** of the load balancer (found in the Description tab).
5. Paste the **DNS name URL** in a browser to verify the site loads and traffic is balanced across instances. It might take few refreshes to make the DNS run correctly.
6.



Step 7: Clean Up (Optional)

- To avoid AWS charges, delete the load balancer, target group, security group, and terminate EC2 instances if no longer needed.